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| **Subject: Programming with Java** | |
| **ASSIGNMENT No : 2** | |
| **TITLE : Constructors in Java** | |

**Problem Statement**

Develop a Java application demonstrating the use of default, parameterized, and copy constructors for object initialization and data management.

**Learning Objectives**

To understand and implement default, parameterized, and copy constructors for object initialization.

**Programme**

```
public class BookDemo {


    String title;

    String author;

    double price;


    BookDemo() {

        title = "Untitled";

        author = "Unknown";

        price = 0.0;

    }


    BookDemo(String t, String a, double p) {
```

```
    title = t;
    author = a;
    price = p;
}

BookDemo(BookDemo b) {
    title = b.title;
    author = b.author;
    price = b.price;
}

void displayInfo() {
    System.out.println("Title  : " + title);
    System.out.println("Author : " + author);
    System.out.println("Price  : Rs." + price);
}

public static void main(String[] args) {

    BookDemo b1 = new BookDemo();
    System.out.println("Book 1 (default constructor):");
    b1.displayInfo();

    System.out.println("----------------------------");

    BookDemo b2 = new BookDemo("Java Basics", "R. Sharma", 350.0);
    System.out.println("Book 2 (parameterized constructor):");
```

```
        b2.displayInfo();

        System.out.println("----------------------------");

        BookDemo b3 = new BookDemo(b2);
        System.out.println("Book 3 (copy of Book 2 using copy constructor):");
        b3.displayInfo();

        System.out.println("----------------------------");

        b3.price = 400.0;
        System.out.println("After changing Book 3's price:");
        System.out.println("Book 2 price: Rs." + b2.price);
        System.out.println("Book 3 price: Rs." + b3.price);
    }
}
```

**Output**

```
Book 1 (default constructor):
Title  : Untitled
Author : Unknown
Price  : Rs.0.0
----------------------------
Book 2 (parameterized constructor):
Title  : Java Basics
Author : R. Sharma
Price  : Rs.350.0
----------------------------
Book 3 (copy of Book 2 using copy constructor):
Title  : Java Basics
Author : R. Sharma
Price  : Rs.350.0
----------------------------
After changing Book 3's price:
Book 2 price: Rs.350.0
Book 3 price: Rs.400.0

=== Code Execution Successful ===
```

**Conclusion**

This program demonstrates default, parameterized, and copy constructors for object initialization in Java. The default constructor sets preset values, the parameterized constructor accepts custom values at creation, and the copy constructor duplicates an existing object independently. The output confirms that modifying the copied object (`b3`) does not affect the original (`b2`), proving each object holds its own separate data in memory.